

- 2D Arrays ($m \times n$)
- Iterations using nested loops
- Row/Col Major
- Solved Problems
- Jagged Array (#diff col)
- Problem (Avg of each row)
- Strings and Array of Strings
- Problems
- String to char array (`s.split("")`)
- Problems

- Time Complexity

$$\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline \end{array}$$

1 2 3 4 5 6 ✓

1 4 7 2 5 8 ✓

...

row $\rightarrow [0][i]$

Jagged Array

—

2-D Array

with different
no. of elements
in each row.

7	8	2	0	
3	1	-2	5	7
6	9			
10	11	1		

```

57 // Problem
58 // you are given a matrix where data of few students is mentioned,
59 // you have to return an array with the average value of each row

```

$i = 0 \rightarrow \text{sum} = 0$

$j: 0 \rightarrow 4$
 $7 + 8 + 2 + 0$

$\text{sum} = 17$

$\text{avg} = 17/4 = 4.25$

Strings

let s = "Hello";

e → s[.]

0 1 2 3 4
H e l l o

Represented via

" "

or

' '

let str = "Hello World";

i = 0 → "

e

"

0

"

0

"

* Convert String into Array of characters

* Array of Strings

2D

s = ["Raj", "Priya", "Kahni"]

0

0

R

a

2

j

3

4

1

P

y

i

h

a

2

K

a

n

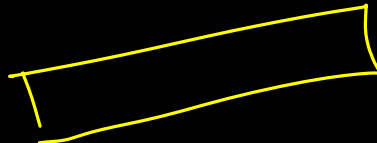
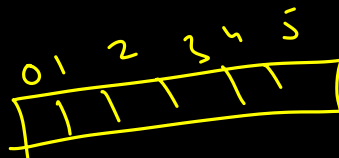
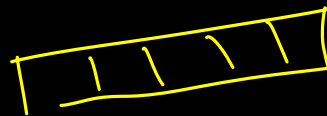
i

1

Time Complexity

$$f(n) = n$$

$$f(n) = \frac{n}{2}$$



~~$f(n, m) = n \times m$~~

```
[for (i = 0 → n) {  
  for (j = 0 → m) {  
    print(i, j)  
  }  
}]
```

$$n \times m$$

$i = 0 \rightarrow (m)$
 $i = 1 \rightarrow (m)$

$i = n-1 \rightarrow m$

$i = 0 \rightarrow 5 \times g(n)$

$j = 1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16$

$\leftarrow 67 \leftarrow 32$
 $\textcircled{128}$

```
let cnt=0;  
for(let i=0; i<5; i++){  
  for(let j=1; j<100; j*=2){  
    cnt++;  
  }  
}  
  
console.log(cnt);
```

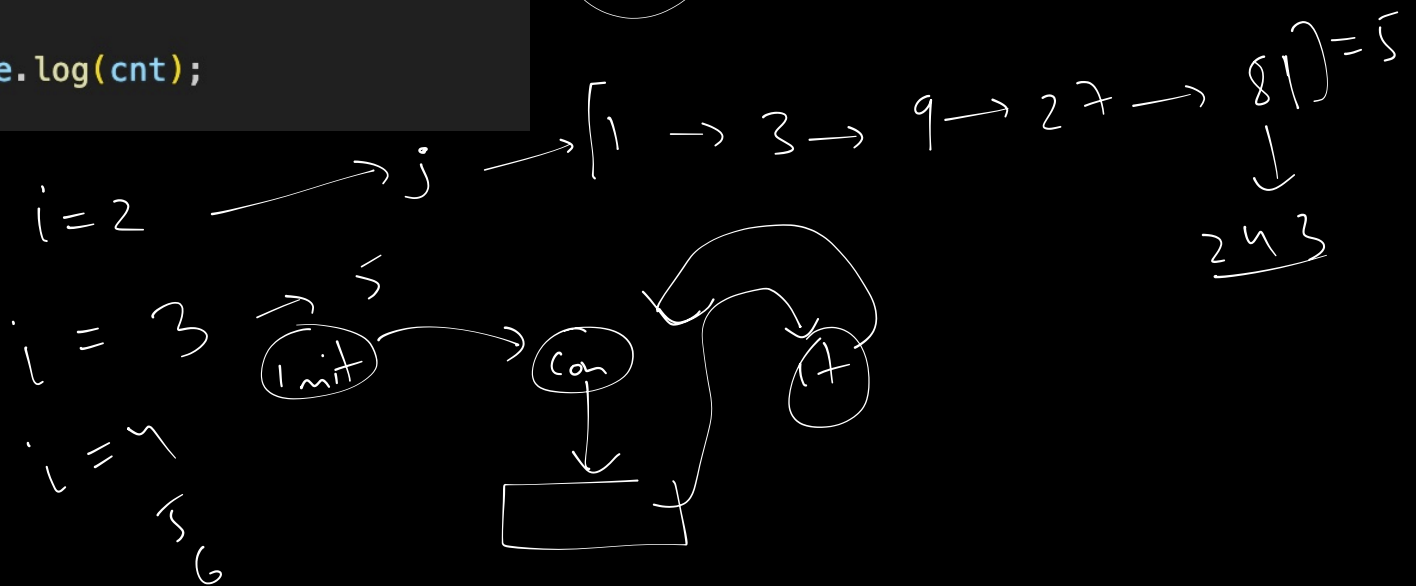
$g(n)$

```

cnt=0;
for(let i=2; i<=6; i++){
  for(let j=1; j<100; j*=3){
    cnt++
  }
}
console.log(cnt);

```

25



```

cnt = 0;
for(let i=0; i<10; i+=2){
  for(let j=100; j>2; j=j/2){
    cnt++;
  }
}

```

$100 \rightarrow 50 \rightarrow 25 \rightarrow 12.5$

6×5

$6 \cdot 25$

$3 \cdot 12.5$

$\downarrow$

$\times \boxed{1. - < 2}$

$i=0 \checkmark$

$i=2$

```
// dependent on outer loop
cnt = 0;
for(let i=0; i<5; i++){
  for(let j=0; j<i; j++){
    cnt++;
  }
}

console.log(cnt);
```

10

$i=0 \rightarrow j=0 \times$ 0
 $i=1 \rightarrow j=0 \rightarrow 1$ 1
 $i=2 \rightarrow j=0 \rightarrow 1 \rightarrow 2$ 2
 $i=3 \rightarrow j=0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ 3
 $i=4 \rightarrow j=0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ 4
 $i=5 \rightarrow \cancel{j=0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5}$

```
// dependent on outer loop
cnt = 0;
for(let i=0; i<n; i++){
  for(let j=0; j<i; j++){
    cnt++;
  }
}

// console.log(cnt);
```

$i=0 \Rightarrow j=0 \times = 0$
 $i=1 \Rightarrow j=0 \rightarrow 1 = 1$
 $i=2 \Rightarrow j=0 \rightarrow 1 \rightarrow 2 = 2$
 $\vdots$
 $i=n-1 \Rightarrow j=0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow n-1 = \underline{\underline{n-1}}$
 $\times (n-1)$

* logarithm

$$b = a^x$$

$$8 = 2^3$$

$\Downarrow$

$$\log_a b$$

$$= x$$

$$\log_2 8 = 3$$

$$* \log b^n = n * \log b$$

|

$$\log$$

eg:-

$$\log 3^9$$

$$\log 7^7$$

$$\log 5^1$$

$$\log 8^8$$

$$\log_2 (1024)$$

$$n = 1024$$

$$\log_2^n \quad \log_2 1024$$

$$\# \underline{\underline{1024}}$$

$$\# \underline{\underline{10}}$$

$$g:- 2^7 = 64$$

$$10^4 = 10000$$

$$\boxed{3^n = 81}$$

$$2^{\textcircled{3}} = \underline{2 \times 2 \times 2}$$

$$9^2 = 729$$

$$5^7 = 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$$

$$a^n = b$$

$$\lfloor \# \log_a(b^n) = n * \log_a b \rfloor^x$$

$$g := \log_2 8^3$$

$$\log_5 625 = 4 \quad \log_b b = 1$$

$$\log_2 2^3$$

$$\log_5 5^7$$

$$\log_2 4^7$$

$$= (2^2)^7 = \log_2 2^{14} = 14$$

```
let cnt1 = 0;
for(let i=100; i>2; i/=2){
  for(let j=0; j<i; j++){
    cnt1++;
  }
}
console.log(cnt1);
```

$$i = \boxed{} < 2 \quad \times$$

$$i = 100$$

$$j = [0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow 99 \rightarrow 100^x]$$

$$i = 50$$

$$\rightarrow 50$$

$$i = 25$$

$$\rightarrow 25$$

$$i = 12.5$$

$$\rightarrow 13$$

$$i = 6.25$$

$$\rightarrow 7$$

$$i = 3.125$$

$$\rightarrow 4$$

```
let cnt2 = 0;
for(let i=0; i<100; i++){
  for(let j=i; j>2; j/=2){
    cnt2++;
  }
}
console.log(cnt2);
```

$$i = 66 \rightarrow (6)$$

$$\downarrow j = [66 \rightarrow 33 \rightarrow 16.5 \rightarrow 8.25 \rightarrow 4.125 \rightarrow 2.0625]$$

$$(1, \dots)$$

0	0
1	
2	
4	1
6	2

$$i = 0 \rightarrow j = 0 \times \mid 2^x \leq 3$$

$$i = 1 \rightarrow j = 1 \times$$

$$i = 2 \rightarrow j = [2^x \rightarrow 2]$$

$$i = 3 \rightarrow j = [3 \rightarrow 1.5] (1)$$

$$i = 4 \rightarrow j = [4 \rightarrow 2 \rightarrow] (1)$$

$$i = 5 \Rightarrow j = [5 \rightarrow 2.5 \rightarrow 1.25] (2)$$

$$(\log_2 i)_{int} - 1$$

$$\frac{16}{3}$$

$$\frac{32}{4}$$

$$\frac{64}{5}$$

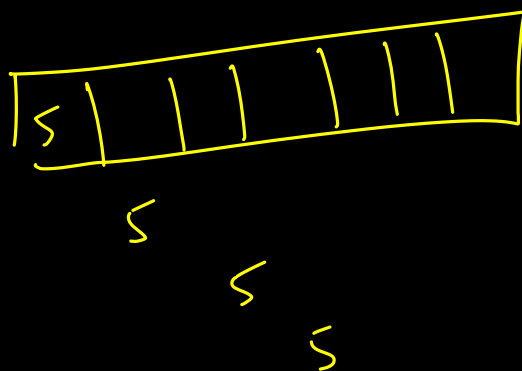
$$\frac{100}{6}$$

$$\frac{128}{7}$$

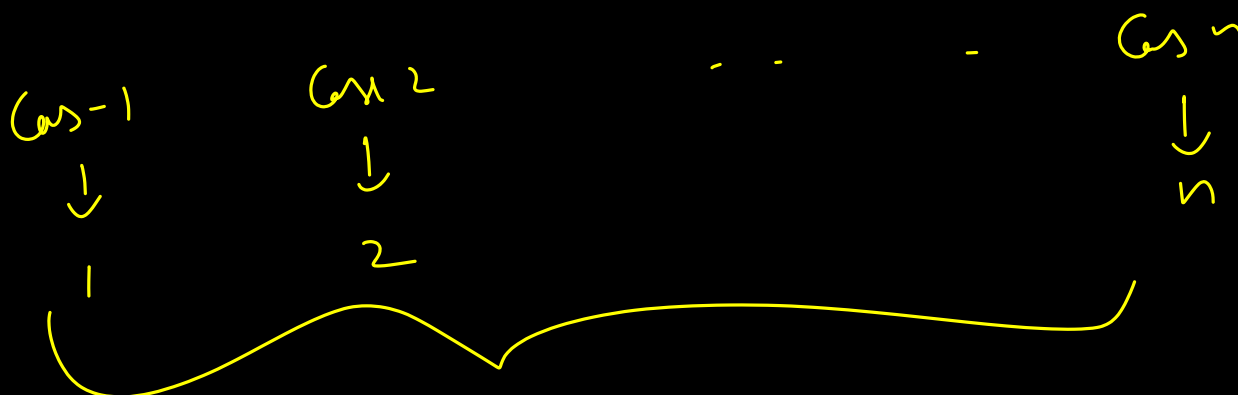
* Best Case $\rightarrow 1$

Avg Case $\rightarrow \frac{n+1}{2}$

Worst Case $\rightarrow n$



Case -



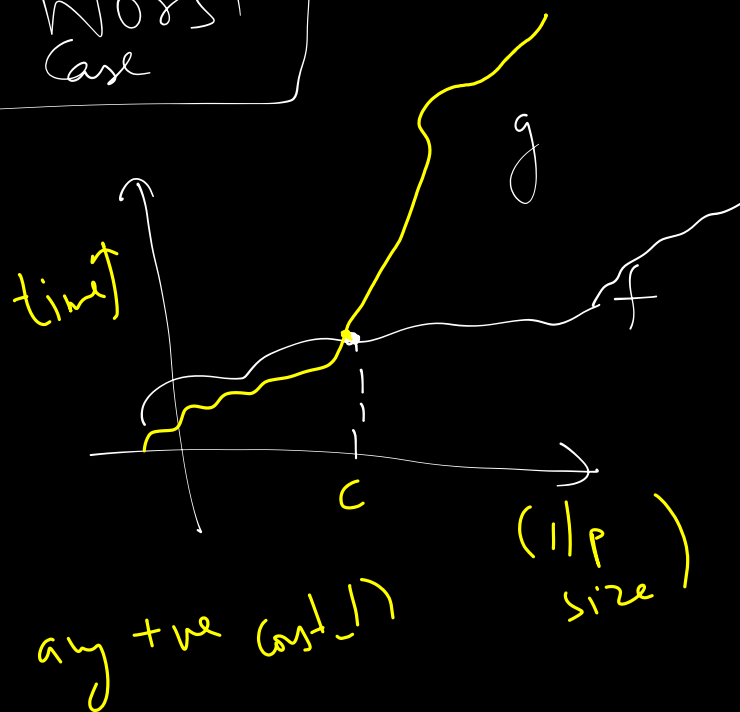
$$\frac{n(n+1)}{2}$$

$$\text{Avg} = \frac{\frac{n(n+1)}{2} \times n}{n} = \underline{\underline{\frac{n+1}{2}}}$$

* Big O

Big O $\neq$ Worst Case

$$f = O(g(n))$$



$$f(n) \leq k \cdot g(n)$$

$$g: g = 2n$$

$$f = 5n$$

$$5n \leq k \cdot 2n$$

$$\underline{f(n) = O(g(n))}$$

$$O(2^n) = O(n)$$

$$1 \longrightarrow \underline{\underline{i \neq 2}}$$

$f(n) \longrightarrow$

n	$\frac{n}{2}$	$\log n$	$\boxed{\cancel{2^n}}$
-----	---------------	----------	------------------------

$$n \approx \frac{n}{2}$$

$$f = n^3 + n^2 + n$$

$O(n^3)$

$$f = 5n^2 + 4n^3$$

$$4n^3$$

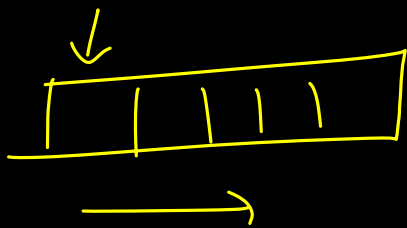
$$\underline{\underline{O(n^3)}}$$

$$f = n + \log n$$

$O(n)$

$$f = n * \log n + n$$

$=$



$\rightarrow O(1)$

$$f = n + n^2$$

$$\underline{\underline{= O(n^2)}}$$

```
[ for ( i = 0 → n ) {  
    }  
]
```

```
[ for ( i = 0 → n )  
    for ( i = 0 → n )  
    }  
]
```

$$f = n + n + n = 3n$$

$$3n$$

code

```
for ( i = 0 → n ) {
    for ( j = 0 →  $\frac{n}{2}$  ) {
```

```
    }
```

```
}
```

```
for ( i = 0 →  $\frac{n}{3}$  ) {
```

```
}
```

$$f = \frac{n^2}{2} + \frac{n}{3}$$

$$= \frac{n^2}{2}$$

$$= O(n^2)$$